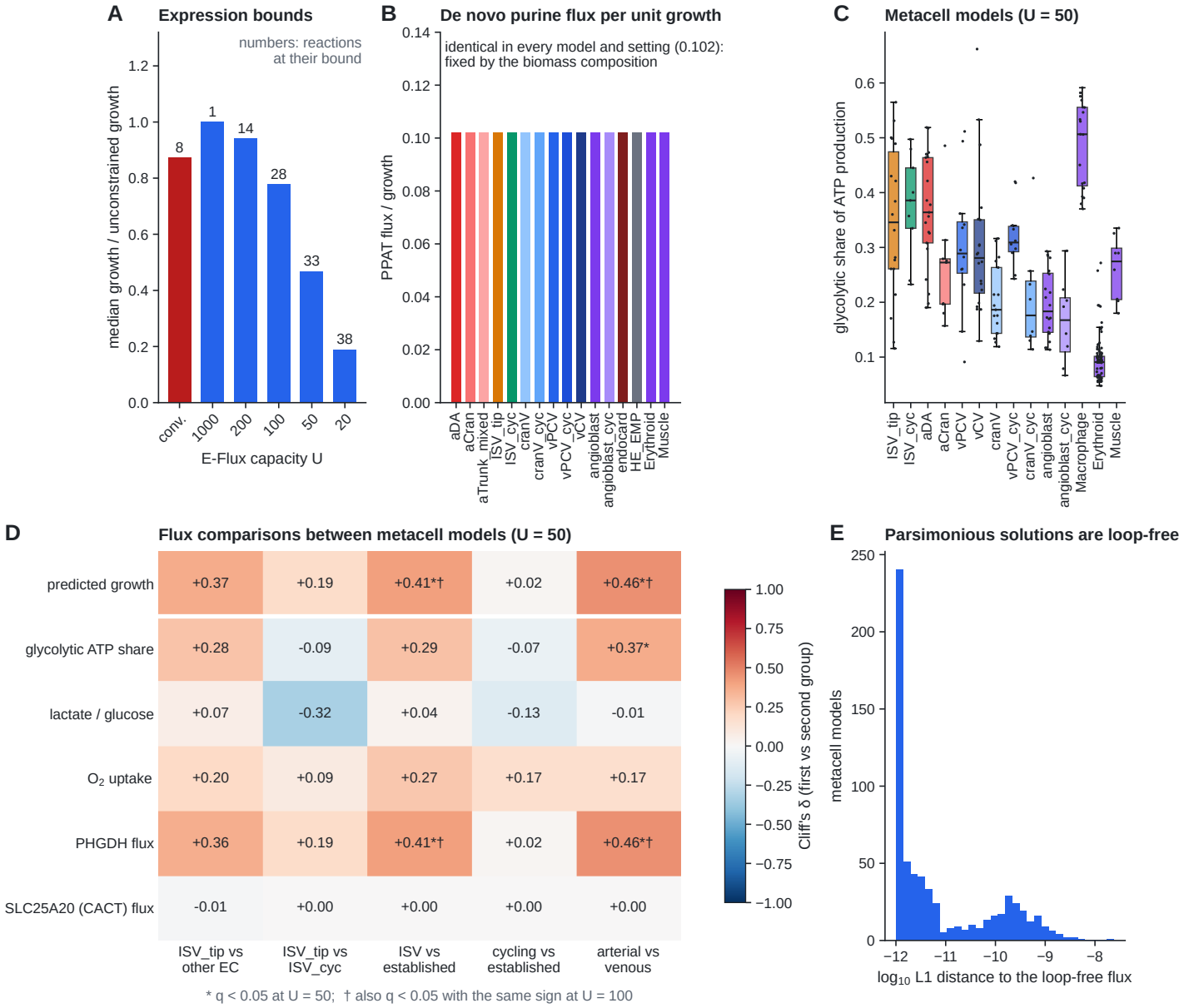

A
Expression bounds
numbers: reactions at their bound
median growth / unconstrained growth
E-Flux capacity U
conv.
1000
200
100
50
20
8
1
14
28
33
38
B
De novo purine flux per unit growth
identical in every model and setting (0.102): fixed by the biomass composition
PPAT flux / growth
aDA
aCran
aTrunk_mixed
ISV_tip
ISV_cyc
cranV
cranV_cyc
vPCV
vPCV_cyc
vCV
angioblast
angioblast_cyc
endocard
HE_EMP
Erythroid
Muscle
C
Metacell models (U = 50)
glycolytic share of ATP production
ISV_tip
ISV_cyc
aDA
aCran
vPCV
vCV
cranV
vPCV_cyc
cranV_cyc
angioblast
angioblast_cyc
Macrophage
Erythroid
Muscle
D
Flux comparisons between metacell models (U = 50)
predicted growth
glycolytic ATP share
lactate / glucose
O2 uptake
PHGDH flux
SLC25A20 (CACT) flux
+0.37
+0.19
+0.41*†
+0.02
+0.46*†
+0.28
-0.09
+0.29
-0.07
+0.37*
+0.07
-0.32
+0.04
-0.13
-0.01
+0.20
+0.09
+0.27
+0.17
+0.17
+0.36
+0.19
+0.41*†
+0.02
+0.46*†
-0.01
+0.00
+0.00
+0.00
+0.00
ISV_tip vs other EC
ISV_tip vs ISV_cyc
ISV vs established
cycling vs established
arterial vs venous
Cliff's δ (first vs second group)
* q < 0.05 at U = 50; † also q < 0.05 with the same sign at U = 100
E
Parsimonious solutions are loop-free
metacell models
log10 L1 distance to the loop-free flux